

ILE D'ORLEANS, QC, Canada

WMO: 714240

Lat: 46.9967N

Lon: 70.8083W

Elev: 4

StdP: 101.28

Time Zone: -5.00 (NAE)

Period: 90-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.8	-20.9	-29.9	0.2	-23.4	-27.1	0.3	-20.0	16.4	-5.6	15.7	-6.7	6.4	230	0.740

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.4	27.9	20.1	26.2	19.2	24.5	18.4	21.8	25.1	20.8	23.9	19.9	23.0	7.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	15.4	23.6	19.7	14.4	22.8	18.7	13.5	21.7	63.8	25.3	60.3	23.9	57.0	23.1	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.3	14.2	12.5		-26.1	30.7	2.3	1.3	-27.8	31.6	-29.1	32.3	-30.4	33.0	-32.1	33.9
			DB	-26.4	23.4	2.0	1.7	-27.9	24.6	-29.1	25.7	-30.2	26.6	-31.7	27.9
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-9.7	-8.6	-2.8	3.7	11.3	17.1	19.3	19.2	14.5	7.8	1.2	-4.9	(6)
(7)		DBStd	11.13	6.22	6.02	4.76	3.43	3.80	3.68	2.74	2.72	3.51	3.70	4.31	5.56	(7)
(8)		HDD10.0	2569	612	521	397	192	27	1	0	0	4	88	267	461	(8)
(9)		HDD18.3	4729	870	754	656	439	220	65	21	22	123	326	515	719	(9)
(10)		CDD10.0	1019	0	0	0	2	68	215	289	284	139	21	1	0	(10)
(11)		CDD18.3	139	0	0	0	0	3	29	51	48	8	0	0	0	(11)
(12)		CDH23.3	615	0	0	0	1	26	134	222	197	34	0	0	0	(12)
(13)		CDH26.7	108	0	0	0	0	5	26	42	31	6	0	0	0	(13)
(14)	Wind	WSAvg	5.7	6.2	5.8	5.9	6.1	6.1	5.5	5.1	5.2	5.4	5.8	5.8	6.0	(14)
(15)	Precipitation	PrecAvg	1144	83	70	70	84	86	113	129	116	102	107	89	94	(15)
(16)		PrecMax	1354	139	119	119	186	128	219	229	185	149	227	130	155	(16)
(17)		PrecMin	891	27	47	24	10	43	47	67	25	56	29	39	43	(17)
(18)		PrecStd	141	33	20	24	46	25	35	41	37	23	48	23	34	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	5.6	8.4	17.7	26.3	29.5	30.2	29.3	27.0	20.6	15.9	6.6	(19)
(20)			MCWB	3.2	3.3	3.8	11.7	17.7	20.4	21.2	20.2	19.7	14.3	13.6	4.2	(20)
(21)		2%	DB	1.9	3.0	5.5	12.7	22.5	27.0	27.6	27.3	23.6	17.2	11.0	4.0	(21)
(22)			MCWB	0.8	1.3	2.8	7.9	14.6	18.8	20.1	20.2	18.3	13.3	8.6	2.3	(22)
(23)		5%	DB	0.2	1.3	3.9	9.7	19.5	24.6	25.8	25.6	21.4	14.5	8.6	2.7	(23)
(24)			MCWB	-0.6	0.1	1.7	5.7	13.5	17.8	19.2	19.4	17.2	11.3	6.2	1.3	(24)
(25)		10%	DB	-1.5	-0.1	2.5	7.9	16.9	22.7	24.0	23.9	19.7	12.7	6.6	1.7	(25)
(26)			MCWB	-2.3	-1.0	0.7	4.7	12.4	17.0	18.6	18.8	15.9	10.0	4.5	0.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	4.3	5.6	13.1	18.9	22.3	23.7	22.5	21.2	17.1	13.9	5.1	(27)
(28)			MCDWB	4.1	5.2	7.3	15.1	25.8	26.6	27.9	26.2	24.2	18.8	16.8	6.1	(28)
(29)		2%	WB	0.9	1.5	3.4	8.4	16.1	20.4	22.2	21.6	19.8	14.0	9.3	2.7	(29)
(30)			MCDWB	1.6	2.7	4.7	11.6	20.4	24.0	25.5	24.9	22.6	16.8	10.4	3.7	(30)
(31)		5%	WB	-0.7	0.3	2.2	6.4	14.5	19.5	20.7	20.9	18.2	12.1	6.4	1.6	(31)
(32)			MCDWB	0.0	1.2	3.7	9.0	18.5	23.0	23.6	23.9	20.8	14.1	8.1	2.5	(32)
(33)		10%	WB	-2.5	-1.1	1.0	5.1	12.7	18.5	19.4	19.9	16.6	10.5	4.8	0.7	(33)
(34)			MCDWB	-1.8	-0.2	2.3	7.3	16.3	22.0	22.4	22.9	19.0	12.2	6.3	1.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.2	7.3	6.1	5.8	7.5	7.6	7.4	7.6	7.0	5.7	5.1	5.8	(35)
(36)			MCDBR	8.9	7.1	7.0	9.5	12.6	10.8	10.1	9.7	9.8	8.6	7.7	5.6	(36)
(37)			MCWBR	8.8	6.4	5.1	6.1	6.8	5.3	4.9	5.0	6.0	6.2	7.0	5.2	(37)
(38)		5% WB	MCDBR	8.6	6.9	6.8	8.4	11.1	9.3	8.3	8.3	8.8	8.0	7.6	5.3	(38)
(39)			MCWBR	8.5	6.3	5.4	6.1	6.8	5.6	5.4	5.8	6.4	6.4	7.3	5.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.262	0.285	0.326	0.357	0.367	0.402	0.417	0.395	0.373	0.341	0.310	0.288	(40)	
(41)		taud	2.338	2.261	2.227	2.331	2.382	2.303	2.285	2.347	2.402	2.479	2.452	2.313	(41)	
(42)		Ebn at Noon	836	883	892	897	897	863	844	849	836	814	771	746	(42)	
(43)		Edh at Noon	72	96	117	117	116	126	127	115	99	79	66	67	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.39	2.30	3.49	4.44	5.06	5.49	5.36	4.66	3.65	2.13	1.23	0.94	(44)	
(45)		RadStd	0.19	0.28	0.28	0.38	0.53	0.35	0.54	0.30	0.30	0.18	0.14	0.09	(45)	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page